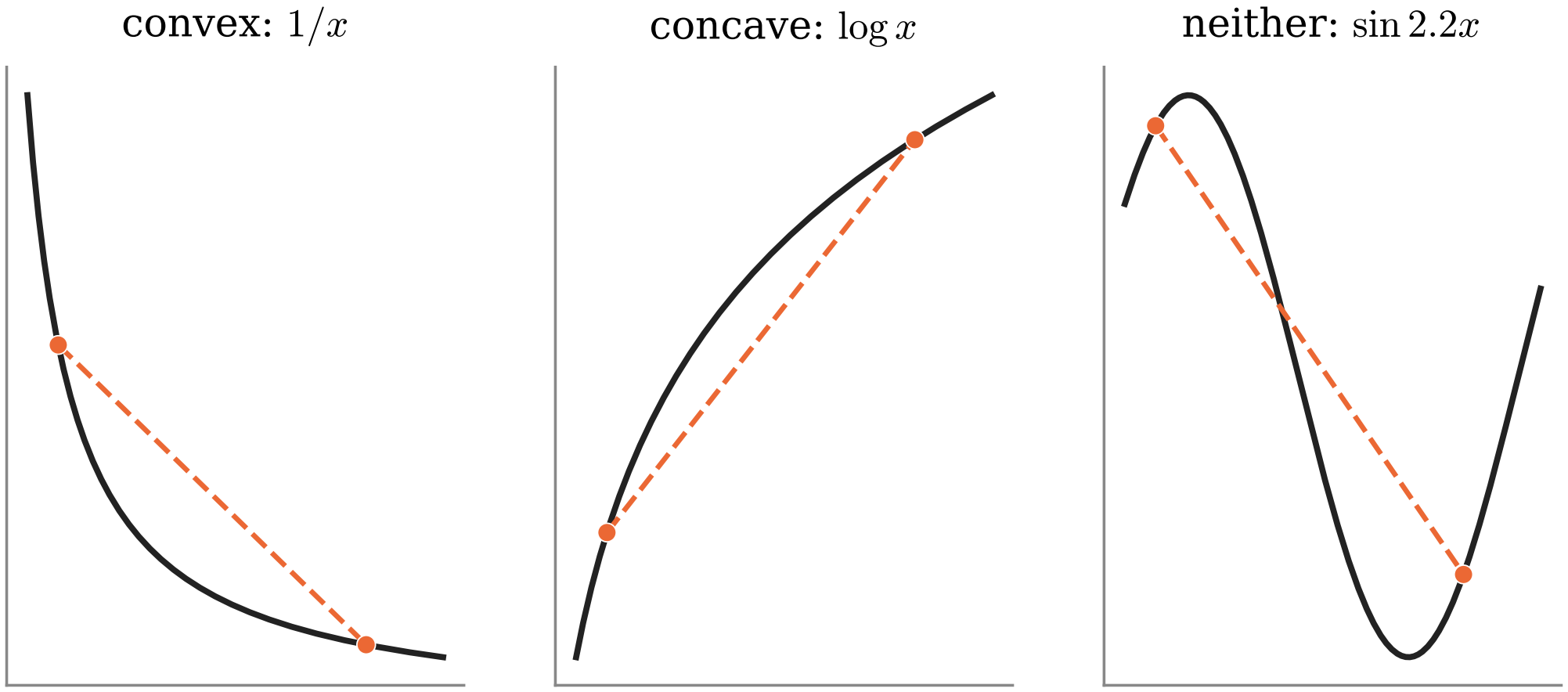
convex: $1/x$
concave: $\log x$
neither: $\sin 2.2x$